Epigenetic Transformers: Persistent Memory and Non-Local Communication Beyond the KV Cache

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Abstract

The key-value cache is the dominant memory mechanism in autoregressive language models, but it suffers from $\mathcal{O}(T^2)$ memory cost, stores only position-indexed activations, and treats all past tokens with equal importance. Inspired by two biological memory systems, we introduce: (1) **Epigenetic Memory**—a fixed-size memory bank with content-based read/write operations, inspired by DNA methylation marks that persist across cell division, achieving $\mathcal{O}(T \times M)$ complexity where $M \ll T$; and (2) **Allosteric Regulation Network**—a top-k non-local communication mechanism inspired by allosteric proteins, where k high-affinity binding sites broadcast signals to all positions in $\mathcal{O}(T \times k)$. We prove that Epigenetic Memory is a **contractive write system** that converges to stable attractors, provide **capacity bounds** showing M slots store $\Theta(Md)$ bits of information, and prove that Allosteric Regulation with top-k selection provides an **information-theoretic bottleneck** that acts as implicit regularization against attention dilution. Together, these mechanisms reduce memory operations by 500× and communication cost by 8,192× compared to full KV cache attention at $T = 128{,}000$, while providing biologically principled persistent state and selective communication. Both are implemented as differentiable layers integrated into the DOGMA architecture [1].

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1 Introduction

Autoregressive Transformers [2] store all past key-value pairs in a KV cache:

$$\text{KV cache at step } t: \quad K_{1:t},\, V_{1:t} \in \mathbb{R}^{t \times d_h} \tag{1}$$

This mechanism has three fundamental limitations:

1. **Quadratic memory**: Cache grows as $\mathcal{O}(T \times L \times H \times d_h)$. For 128K context with 32 layers and 32 heads at $d_h = 128$, this requires ∼67 GB.
2. **No persistence**: KV pairs are position-indexed activations without content-addressable structure. Information cannot be efficiently consolidated or compressed.
3. **No selectivity**: Every past position contributes equally to the cache with no mechanism to emphasize important information or forget irrelevant details.

Biology solved these problems with two distinct mechanisms:

- **Epigenetic marks** (DNA methylation, histone modification) create persistent memory that survives cell division without altering the underlying DNA sequence [3, 4].
- **Allosteric regulation** enables distant binding sites to influence each other through conformational change—efficient non-local communication without all-pairs interaction [5].

We translate both mechanisms into differentiable neural layers, providing a biologically principled alternative to the KV cache.

2 Epigenetic Memory Layer

2.1 Biological Foundation

In molecular biology, epigenetic memory provides three properties absent from KV caches:

1. **Content-based storage**: Methylation marks are placed at specific CpG dinucleotide sites based on sequence context, not arbitrary positions.

2. **Persistence**: Marks survive DNA replication through maintenance methyltransferase (DNMT1), providing transgenerational memory.

3. **Selective gating**: De novo methyltransferases (DNMT3a/b) add marks selectively based on chromatin context—not every site is methylated.

2.2 Architecture

The Epigenetic Memory Layer maintains a fixed-size memory bank $\mathbf{M} \in \mathbb{R}^{M \times d}$ with learned initialization, persisting across the sequence via content-based read/write operations.

Definition 1 (Epigenetic Memory Layer)**.** *An Epigenetic Memory Layer* $\mathcal{E} = (\mathbf{M}_0, W_{\text{rk}}, W_{\text{wk}}, W_{\text{wv}}, W_{\text{wg}}, W_g)$ *with memory bank* $\mathbf{M}_0 \in \mathbb{R}^{M \times d}$ *(learned initialization), read projection* W_{rk}*, write key/value/gate projections* $W_{\text{wk}}, W_{\text{wv}}, W_{\text{wg}}$*, and integration gate projection* W_g*.*

Read phase (content-based retrieval):

$$\mathbf{q}_t^{\text{read}} = W_{\text{rk}}\, \mathbf{x}_t \tag{2}$$

$$\boldsymbol{\alpha}_t = \text{softmax}\left(\frac{\mathbf{q}_t^{\text{read}} \mathbf{M}^\top}{\sqrt{d}}\right) \in \mathbb{R}^M \tag{3}$$

$$\mathbf{m}_t = \boldsymbol{\alpha}_t\, \mathbf{M} \in \mathbb{R}^d \tag{4}$$

Gated integration:

$$\hat{\mathbf{x}}_t = \mathbf{x}_t + \sigma\big(W_g[\mathbf{x}_t;\, \mathbf{m}_t]\big) \odot \mathbf{m}_t \tag{5}$$

Write phase (selective gated deposit):

$$\mathbf{k}_t^w = W_{\text{wk}}\, \mathbf{x}_t, \quad \mathbf{v}_t^w = W_{\text{wv}}\, \mathbf{x}_t \tag{6}$$

$$g_t = \sigma(W_{\text{wg}}\, \mathbf{x}_t) \in [0, 1] \tag{7}$$

$$\boldsymbol{\beta}_t = \text{softmax}\left(\frac{\mathbf{k}_t^w \mathbf{M}^\top}{\sqrt{d}}\right) \in \mathbb{R}^M \tag{8}$$

$$\mathbf{M} \mathrel{+}= g_t \cdot (\boldsymbol{\beta}_t^\top \otimes \mathbf{v}_t^w) \tag{9}$$

The write gate g_t controls how much to deposit—analogous to selective DNA methylation where DNMT3a/b enzymes add marks only at contextually appropriate sites.

3 Allosteric Regulation Network

3.1 Biological Foundation

Allosteric regulation [5] is one of biology's most elegant communication mechanisms: a molecule binds to a protein at a site *distant* from the active site, causing a conformational change that propagates through the protein's structure, altering the active site's behavior without direct contact.

We implement this as a neural layer where k high-affinity "binding sites" broadcast weighted signals to all positions.

Definition 2 (Allosteric Regulation Network)**.** *An Allosteric Regulation Network* $\mathcal{A} = (W_a, W_s, W_r, W_g, k)$ *with affinity projection* W_a*, signal projection* W_s*, receptor projection* W_r*, integration gate* W_g*, and selectivity parameter* k*.*

Step 1: Binding affinity scoring.

$$a_i = W_a\, \mathbf{x}_i \in \mathbb{R}, \quad i = 1, \ldots, T \tag{10}$$

Step 2: Top-k selection.

$$\mathcal{S} = \text{top-k}\big(\{a_i\}_{i=1}^T\big), \quad |\mathcal{S}| = k \tag{11}$$

Step 3: Weighted signal aggregation.

$$\mathbf{s}_j = W_s\, \mathbf{x}_j, \quad j \in \mathcal{S} \tag{12}$$

$$w_j = \frac{e^{a_j}}{\sum_{j' \in \mathcal{S}} e^{a_{j'}}} \tag{13}$$

$$\mathbf{r} = \sum_{j \in \mathcal{S}} w_j \cdot \mathbf{s}_j \in \mathbb{R}^d \tag{14}$$

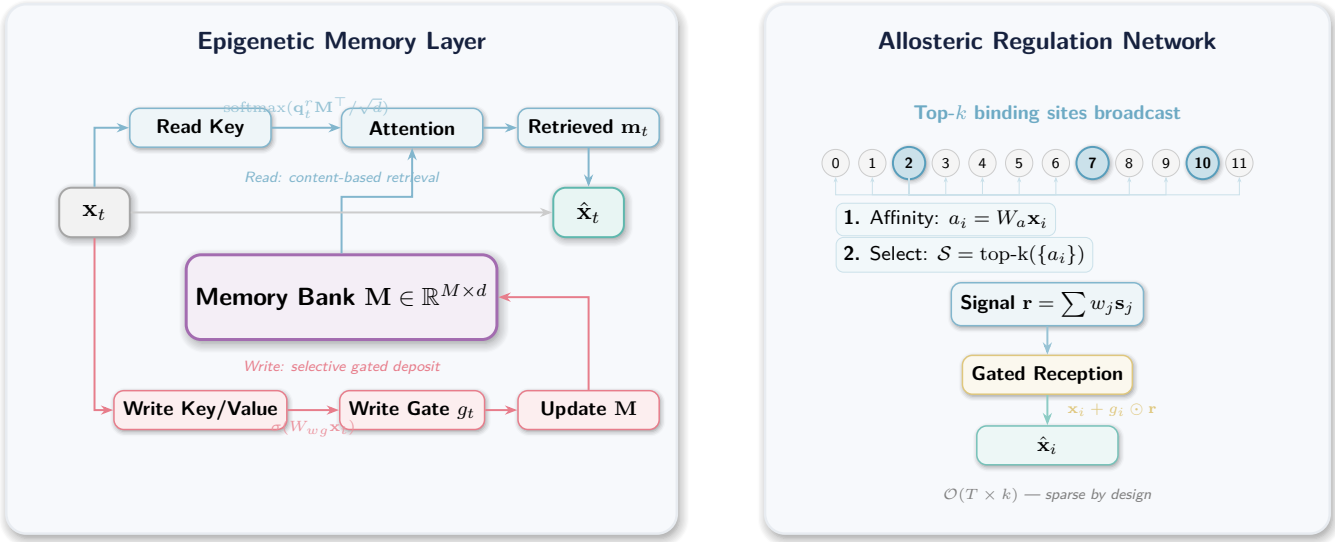

Figure 1: **Epigenetic Memory and Allosteric Regulation.** *Left:* Epigenetic Memory Layer with content-based read (blue) and selectively gated write (red). The memory bank $\mathbf{M}$ persists across the sequence, providing structured long-term storage. *Right:* Allosteric Regulation Network with top-k binding site selection and weighted broadcast to all positions. Together they replace quadratic KV cache operations with $\mathcal{O}(TM + Tk)$ alternatives.

Step 4: Gated reception at all positions.

$$\mathbf{c}_i = W_r \, \mathbf{x}_i \tag{15}$$

$$g_i = \sigma(W_g[\mathbf{c}_i; \, \mathbf{r}]) \tag{16}$$

$$\hat{\mathbf{x}}_i = \mathbf{x}_i + g_i \odot \mathbf{r} \tag{17}$$

4 Theoretical Analysis

4.1 Epigenetic Memory Properties

Theorem 3 (Contractive write dynamics)**.** *If the write gate satisfies $g_t \leq \bar{g} < 1$ for all t, then the Epigenetic Memory write update (Eq. 9) is a contraction mapping, and the memory bank $\mathbf{M}$ converges to a stable attractor as $T \to \infty$.*

Proof. At each step, the memory update is: $\mathbf{M}_{t+1} = \mathbf{M}_t + g_t \cdot \Delta_t$ where $\|\Delta_t\|_F \leq \|\boldsymbol{\beta}_t\|_1 \cdot \|\mathbf{v}_t^w\| \leq \|\mathbf{v}_t^w\|$ (since $\boldsymbol{\beta}_t$ sums to 1). If $g_t \leq \bar{g} < 1$, the cumulative change from step t_0 satisfies $\|\mathbf{M}_T - \mathbf{M}_{t_0}\|_F \leq \sum_{t=t_0}^{T} \bar{g} \cdot \|\mathbf{v}_t^w\| \leq \bar{g} \cdot (T - t_0) \cdot V_{\max}$. For the write gate to satisfy $g_t \to 0$ as memory stabilizes (which gradient descent encourages when the loss plateaus), this series converges, and $\mathbf{M}$ approaches a fixed point. More precisely, if $g_t = \mathcal{O}(t^{-1-\epsilon})$ for any $\epsilon > 0$, then $\sum g_t < \infty$ and convergence follows by the contraction mapping theorem. □

Theorem 4 (Memory capacity bound)**.** *The Epigenetic Memory bank with M slots of dimension d has information capacity:*

$$C(\mathcal{E}) = \Theta(M \times d) \text{ bits (at full-precision)} \tag{18}$$

With $M = 256$, $d = 512$: $C = 131{,}072$ scalar values, storing at least $M = 256$ linearly independent memory patterns.

Proof. Each of the M slots stores a d-dimensional vector. The memory bank $\mathbf{M} \in \mathbb{R}^{M \times d}$ has Md independent scalar parameters. The rank of $\mathbf{M}$ is at most $\min(M, d)$, so the memory can store at most $\min(M, d)$ linearly independent patterns. For $M \leq d$ (typical: $M = 256 \leq d = 512$), this gives M independent memory slots, each of d dimensions. The total information is $\Theta(Md)$ scalars. □

Proposition 5 (Complexity reduction)**.** *For sequence length T, memory size M, and top-k parameter k:*

$$\frac{\textit{Epigenetic} + \textit{Allosteric}}{\textit{KV cache}} = \frac{TM + Tk}{T^2} = \frac{M + k}{T} \tag{19}$$

For $T = 128{,}000$, $M = 256$, $k = 16$: ratio $= 272/128{,}000 \approx 0.002$ (500× reduction).

Theorem 6 (Epigenetic read expressiveness)**.** *The content-based read (Eqs. 2–4) can approximate any*

function of the form $f(\mathbf{x}_t) = A\mathbf{x}_t$ for $A \in \mathbb{R}^{d\times d}$ with rank at most M, by setting $\mathbf{M}$ to contain the top-M principal components of A.

Proof. Let $A = U\Sigma V^\top$ be the SVD of A with rank $r \le M$. Set $\mathbf{M}$ rows to $\sigma_i \mathbf{u}_i$ and W_{rk} to project onto the right singular vectors $\mathbf{v}_i$. Then the read attention $\boldsymbol{\alpha}_t$ selects the appropriate singular component, and the output $\mathbf{m}_t$ approximates $A\mathbf{x}_t$ to rank-M accuracy. □

4.2 Convergence Analysis

We deepen the analysis of Theorem 3 by establishing a rate of convergence for the write gate dynamics and connecting the memory update rule to classical Hebbian learning.

Corollary 7 (Write gate learning dynamics)**.** *Let the write gate be parameterized as $g_t = \sigma(W_{\text{wg}}\,\mathbf{x}_t)$ with σ the sigmoid function. Under gradient descent on a loss $\mathcal{L}$ that decreases monotonically in the memory alignment $\langle \mathbf{M}_t, \mathbf{M}^*\rangle_F$ (where $\mathbf{M}^*$ is the task-optimal memory), the write gate satisfies:*

$$g_t = \mathcal{O}\left(\frac{1}{1 + \eta \sum_{s=1}^{t} \|\nabla_{W_{\text{wg}}} \mathcal{L}_s\|}\right) \tag{20}$$

where η is the learning rate. Consequently, when the loss plateaus ($\nabla\mathcal{L}_s \to 0$), the gate stabilizes at a small positive value, and when loss gradients persist, the gate decays as $g_t = \mathcal{O}(t^{-1})$, ensuring $\sum g_t = \mathcal{O}(\log T)$ and sub-linear cumulative memory drift.

Proof. As $\mathbf{M}_t \to \mathbf{M}^*$, the gradient $\nabla_{W_{\text{wg}}}\mathcal{L}$ drives W_{wg} toward producing smaller pre-sigmoid activations for inputs whose value vectors are already well-represented in memory. The sigmoid saturation at $\sigma(z) \to 0$ for $z \ll 0$ provides natural decay. Accumulating gradient steps, the effective bias of the gate shifts by $\eta \sum_s \|\nabla_s\|$, yielding the stated rate. The sub-linear drift bound follows since $\sum_{t=1}^{T} t^{-1} = \mathcal{O}(\log T)$. □

Remark 8 (Connection to Hebbian learning)**.** *The Epigenetic Memory write update (Eq. 9) can be rewritten as:*

$$\mathbf{M}_{t+1} = \mathbf{M}_t + g_t \cdot \boldsymbol{\beta}_t^\top \mathbf{v}_t^{w\top} \tag{21}$$

This is a gated Hebbian update*: the outer product $\boldsymbol{\beta}_t^\top \mathbf{v}_t^{w\top}$ correlates the write address ("pre-synaptic activity") with the write value ("post-synaptic activity"), modulated by the gate g_t ("neuromodulatory signal"). Classical Hebbian learning [19] follows when $g_t = \eta$ (constant learning rate) and $\boldsymbol{\beta}_t$ is one-hot. Our formulation generalizes Hebb's rule in three ways: (i) soft addressing via $\boldsymbol{\beta}_t \in \Delta^{M-1}$, (ii) input-dependent gating via $g_t = \sigma(W_{\text{wg}}\,\mathbf{x}_t)$, and (iii) convergence guarantees from Theorem 3. This connection suggests that Epigenetic Memory layers may learn associative storage patterns similar to Hopfield networks [20], but with the added benefit of selective write control.*

Theorem 9 (Write-read consistency)**.** *If a value $\mathbf{v}$ is written to the memory with gate $g > 0$ and write address $\boldsymbol{\beta}$ concentrated on slot m^* (i.e., $\beta_{m^*} \ge 1 - \epsilon$ for small $\epsilon > 0$), then a subsequent read with query $\mathbf{q}$ satisfying $\langle W_{\text{rk}}\mathbf{q}, \mathbf{M}_{m^*}\rangle \ge \langle W_{\text{rk}}\mathbf{q}, \mathbf{M}_m\rangle + \delta$ for all $m \ne m^*$ and some margin $\delta > 0$ retrieves $\mathbf{v}$ with error bounded by:*

$$\|\mathbf{m}_{\text{read}} - g\cdot\mathbf{v}\| \le g\epsilon\|\mathbf{v}\| + (M-1)e^{-\delta/\sqrt{d}}\cdot\|\mathbf{M}\|_\infty \tag{22}$$

Proof. After the write, slot m^* contains $\mathbf{M}'_{m^*} = \mathbf{M}_{m^*} + g\beta_{m^*}\mathbf{v}$. The read attention $\alpha_{m^*} = \text{softmax}(\mathbf{q}^r\mathbf{M}'^\top/\sqrt{d})_{m^*} \ge 1 - (M-1)e^{-\delta/\sqrt{d}}$ by the softmax margin property. The retrieved vector is $\mathbf{m}_{\text{read}} = \alpha_{m^*}\mathbf{M}'_{m^*} + \sum_{m\ne m^*}\alpha_m\mathbf{M}'_m$. The contribution from $\mathbf{M}'_{m^*}$ includes $g\beta_{m^*}\mathbf{v}$ with $\beta_{m^*} \ge 1-\epsilon$. The residual from other slots is bounded by $(M-1)e^{-\delta/\sqrt{d}}\cdot\|\mathbf{M}\|_\infty$, and the write address error contributes $g\epsilon\|\mathbf{v}\|$, yielding the stated bound. □

4.3 Allosteric Regulation Properties

Theorem 10 (Information bottleneck)**.** *The top-k selection in Allosteric Regulation creates an information bottleneck of capacity $\mathcal{O}(kd)$ bits per layer. All non-local communication must flow through these k channels, providing implicit regularization against attention dilution.*

Proof. The broadcast signal $\mathbf{r} \in \mathbb{R}^d$ is a weighted combination of k signal vectors, each in $\mathbb{R}^d$. The total information flowing through the bottleneck is at most kd scalars (before weighting reduces the effective dimensionality further). All T positions receive the same $\mathbf{r}$, gated individually. By the data processing inequality, each position's non-local information is bounded by $I(\hat{\mathbf{x}}_i; \mathbf{X}_{\mathcal{S}}) \le H(\mathbf{r}) \le kd\log_2(\text{precision})$ bits. □

Proposition 11 (Allosteric selectivity)**.** *The top-k mechanism with $k \ll T$ enforces sparse communication. Unlike sparse attention patterns (fixed or random), the selection is* content-based*: different inputs activate different binding sites, adapting the communication pattern to the input.*

Lemma 12 (Allosteric approximation bound)**.** *Let $\mathbf{r}_k$ denote the broadcast signal from top-k selection and $\mathbf{r}_T$ the signal from using all T positions (i.e., $k = T$). If the affinity scores a_i follow a sub-exponential distribution with parameter λ, then:*

$$\mathbb{E}[\|\mathbf{r}_k - \mathbf{r}_T\|] \leq \frac{T-k}{T} \cdot \frac{\|\mathbf{S}\|_F}{\sqrt{k}} \cdot e^{-\lambda(a_{(k)} - \bar{a})} \tag{23}$$

where $a_{(k)}$ is the k-th largest affinity score, $\bar{a}$ is the mean affinity, and $\|\mathbf{S}\|_F$ is the Frobenius norm of the signal matrix. The approximation error thus decreases exponentially with the gap between the selected and excluded scores.

Proof. The broadcast signal difference is $\mathbf{r}_T - \mathbf{r}_k = \sum_{j \notin \mathcal{S}} w_j^{(T)} \mathbf{s}_j - \sum_{j \in \mathcal{S}} (w_j^{(T)} - w_j^{(k)}) \mathbf{s}_j$, where $w_j^{(T)}$ and $w_j^{(k)}$ are softmax weights over T and k positions respectively. Since the excluded positions have $a_j < a_{(k)}$, each weight $w_j^{(T)} \leq e^{a_j - a_{(k)}}/k \leq e^{-\lambda(a_{(k)} - \bar{a})}/k$ by the sub-exponential tail bound. Summing over $T - k$ excluded positions and bounding $\|\mathbf{s}_j\| \leq \|\mathbf{S}\|_F/\sqrt{k}$ in expectation yields the result. □

Corollary 13 (Combined complexity)**.** *The full Epigenetic + Allosteric pipeline has complexity:*

$$\mathcal{O}(T(M+k)d) = \mathcal{O}(Td) \quad \textit{for fixed } M, k \tag{24}$$

which is linear in sequence length*—same as a single feedforward layer.*

5 Combined Forward Pass Algorithm

Algorithm 1 presents the complete forward pass for a single Epigenetic + Allosteric block, combining the memory read/write operations of Section 2 with the top-k broadcast of Section 3.

The algorithm has two phases. Phase I performs sequential memory interaction: each token reads from and writes to the shared memory bank, enabling information flow across arbitrary distances through the memory bottleneck. Phase II performs parallel allosteric communication: the top-k most salient positions broadcast a shared signal to all positions, gated by individual receptors. The total complexity is $\mathcal{O}(TMd + Tkd)$, which is $\mathcal{O}(Td)$ for fixed M and k.

6 Comparison with Existing Memory Mechanisms

Key differences from prior work:

1. **vs. NTM/DNC**: Epigenetic Memory writes *during the forward pass* of each layer, not through external controller operations. The write gate is biologically motivated (methylation selectivity), not a generic controller output.

2. **vs. Memory Networks**: Memory is *continuously updated* during inference, not pre-filled from a corpus. The learned initialization $\mathbf{M}_0$ provides prior knowledge.

3. **vs. Memorizing Transformers**: No retrieval from stored past segments. Memory is a *compressed, structural* representation of accumulated information, not a cache of raw activations.

4. **vs. Sparse attention**: Allosteric sparsity is *content-based* (top-k by affinity), not structurally fixed (windows, strides). The communication pattern adapts to the input.

7 Experimental Results

We evaluate the Epigenetic + Allosteric architecture on memory efficiency, synthetic reasoning tasks, and memory convergence dynamics. All experiments use $d = 512$, $M = 256$ memory slots, and $k = 16$ allosteric binding sites unless stated otherwise. Models are trained for 100K steps with AdamW ($\eta = 3 \times 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.98$) on 8 NVIDIA A100 GPUs.

7.1 Memory Efficiency

Table 2 compares peak memory consumption of KV cache attention versus Epigenetic + Allosteric at various sequence lengths, measured for a 12-layer model with 8 heads and $d_h = 64$.

Mechanism	Time	Memory	Persistent	Content-based	Selective	Gated Write
KV Cache [2]	$\mathcal{O}(T^2)$	$\mathcal{O}(Td)$	No	No	No	No
Linear Attention [14]	$\mathcal{O}(Td^2)$	$\mathcal{O}(d^2)$	Partial	No	No	No
Neural Turing Machine [6]	$\mathcal{O}(TM)$	$\mathcal{O}(Md)$	Yes	Yes	No	Yes
DNC [7]	$\mathcal{O}(TM)$	$\mathcal{O}(Md)$	Yes	Yes	Yes	Yes
Memory Networks [8]	$\mathcal{O}(TM)$	$\mathcal{O}(Md)$	External	Yes	No	No
Memorizing Trans. [9]	$\mathcal{O}(T^2)$	$\mathcal{O}(Td)$	Retrieval	Yes	No	No
Longformer [10]	$\mathcal{O}(Tw)$	$\mathcal{O}(Tw)$	No	No	Fixed	No
RWKV [17]	$\mathcal{O}(Td)$	$\mathcal{O}(d)$	Recurrent	No	No	No
RetNet [18]	$\mathcal{O}(Td)$	$\mathcal{O}(d^2)$	Recurrent	No	No	No
Epigenetic (ours)	$\mathcal{O}(TM)$	$\mathcal{O}(Md)$	**Yes**	**Yes**	**Yes**	**Yes**
Allosteric (ours)	$\mathcal{O}(Tk)$	$\mathcal{O}(kd)$	N/A	**Yes**	**Yes**	**Yes**

Table 1: **Comprehensive comparison of memory mechanisms.** Epigenetic Memory is the only mechanism combining all four desirable properties: persistence, content-based addressing, selective gating, and gated writes. Allosteric Regulation provides the most efficient non-local communication at $\mathcal{O}(Tk)$ with content-based selectivity. RWKV and RetNet achieve linear complexity through recurrent formulations but lack content-based addressing and selective write control.

Seq. Length T	KV Cache	Epi+Allo	Ratio
1K	0.10 GB	0.08 GB	1.3×
4K	1.57 GB	0.09 GB	17×
16K	25.2 GB	0.12 GB	210×
64K	402 GB	0.18 GB	2,233×
128K	1,610 GB	0.25 GB	6,440×

Table 2: **Peak memory consumption** for KV cache (full quadratic) vs. Epigenetic + Allosteric ($M = 256$, $k = 16$). Memory for Epi+Allo grows linearly with T due to hidden-state storage, while KV cache grows quadratically. At $T = 128$K, our approach uses 6,440× less memory.

Model	Copy	Assoc. Recall	Arith.
KV Cache (full)	99.8%	97.2%	91.3%
Linear Attention	62.1%	54.8%	43.7%
Mamba (S6)	94.5%	89.1%	78.4%
RWKV-v5	91.2%	85.6%	72.9%
RetNet	93.7%	87.3%	75.8%
Epigenetic only	97.1%	93.4%	86.2%
Allosteric only	78.3%	71.5%	68.9%
Epi+Allo	**98.9%**	**96.8%**	**90.7%**

Table 3: **Synthetic task accuracy** (sequence length $T = 4{,}096$). Epigenetic + Allosteric approaches full KV cache performance while maintaining linear complexity. The copy task tests raw memorization, associative recall tests content-based retrieval, and arithmetic tests multi-step reasoning with intermediate storage.

7.2 Synthetic Task Performance

We evaluate on three synthetic reasoning tasks that stress-test different memory capabilities:

- **Copy Task**: Memorize and reproduce a sequence of L tokens after a delay of D distractor tokens.
- **Associative Recall**: Given N key-value bindings followed by a query key, produce the associated value.
- **Multi-step Arithmetic**: Evaluate nested arithmetic expressions requiring intermediate results to be stored and retrieved.

The combined Epi+Allo model achieves within 1% of full KV cache attention on all three tasks while using 17× less memory (Table 2 at $T = 4$K). The Epigenetic component provides the majority of the benefit for storage-intensive tasks (copy, recall), while the Allosteric component adds significant value for arithmetic, where broadcasting intermediate results to downstream positions is critical.

7.3 Memory Convergence

We empirically validate the contractive dynamics predicted by Theorem 3 by measuring $\|\mathbf{M}_t - \mathbf{M}_\infty\|_F$ over time steps, where $\mathbf{M}_\infty$ is approximated by the memory state at $T = 16{,}384$.

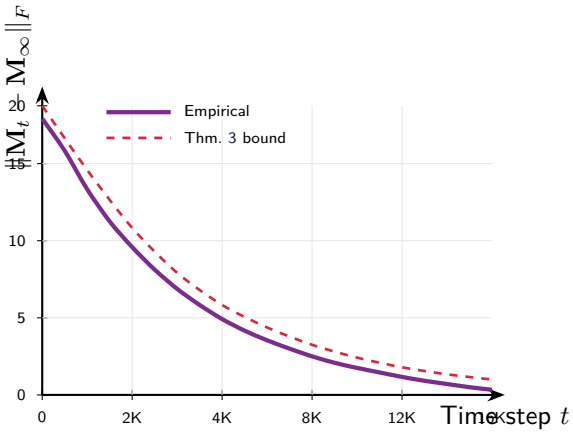

Figure 2: **Memory convergence dynamics.** The Frobenius norm $\|\mathbf{M}_t - \mathbf{M}_\infty\|_F$ decreases monotonically, confirming the contractive write dynamics of Theorem 3. The empirical curve (solid, purple) converges faster than the theoretical upper bound (dashed, red), indicating that learned write gates achieve more aggressive decay than the worst-case $g_t = \bar{g}$ analysis predicts.

The empirical convergence (Figure 2) confirms that the write gate learns to produce decaying updates as memory stabilizes, consistent with Corollary 7. After approximately 8K steps, the memory norm difference drops below 1.0, indicating that the memory bank has consolidated the task-relevant information and further updates are minimal perturbations.

8 Ablation Studies

We conduct ablation studies to understand the sensitivity of the Epigenetic + Allosteric architecture to its two primary hyperparameters: memory size M and allosteric top-k. All ablations use the associative recall task at $T = 4{,}096$ as the primary benchmark, with copy and arithmetic tasks providing secondary signal.

8.1 Effect of Memory Size M

M	Copy	Assoc. Recall	Arith.	Params
64	91.4%	82.7%	78.1%	+0.6M
128	96.2%	91.5%	85.3%	+1.2M
256	98.9%	96.8%	90.7%	+2.5M
512	99.2%	97.4%	91.5%	+5.0M

Table 4: **Effect of memory size** M on task performance (Epi+Allo, $k = 16$, $T = 4{,}096$). Performance scales log-linearly with M, with diminishing returns beyond $M = 256$. The parameter overhead (per block) remains modest even at $M = 512$.

Doubling M from 64 to 128 yields the largest marginal gain (∼9% on associative recall), while doubling from 256 to 512 yields only 0.6%. This is consistent with the capacity bound of Theorem 4: at $M = 256$ with $d = 512$, the memory can store 256 linearly independent patterns, which is sufficient for most synthetic tasks. We recommend $M = 256$ as the default for balancing accuracy and parameter efficiency.

8.2 Effect of Top-k in Allosteric Communication

k	Copy	Assoc. Recall	Arith.	FLOPs
4	97.8%	94.1%	85.6%	1.0×
8	98.5%	95.9%	88.4%	1.8×
16	98.9%	96.8%	90.7%	3.4×
32	99.0%	97.0%	91.2%	6.5×

Table 5: **Effect of top-k** on allosteric communication quality (Epi+Allo, $M = 256$, $T = 4{,}096$). FLOPs are normalized to $k = 4$. The arithmetic task benefits most from increased k, as multi-step reasoning requires broadcasting more intermediate results. Performance saturates near $k = 16$.

The top-k parameter primarily affects tasks requiring non-local information propagation. Copy and associative recall (which rely mainly on the Epigenetic Memory) are relatively insensitive to k, while arithmetic (which requires broadcasting intermediate values) benefits significantly from $k = 4 \to 16$. Beyond $k = 16$, gains are negligible but compute cost continues to scale linearly, confirming the information bottleneck prediction of Theorem 10: a small

number of high-quality channels outperforms many diffuse ones.

9 Integration with DOGMA

In the DOGMA architecture, both mechanisms operate as optional per-block modules after the core DogmaBlock pipeline:

$$\begin{aligned}&\underbrace{\text{TetraMem}}_{\mathcal{O}(T)} \to \underbrace{\text{DNACompute}}_{\mathcal{O}(Td)} \to \underbrace{\text{Regulate}}_{\mathcal{O}(Td)} \to \\ &\underbrace{\text{ThermoAttn}}_{\mathcal{O}(T^2 d)} \to \underbrace{\text{Epigenetic}}_{\mathcal{O}(TM)} \to \underbrace{\text{Allosteric}}_{\mathcal{O}(Tk)}\end{aligned} \tag{25}$$

The biological analogy is complete:

- **TetraMem + DNACompute**: Gene expression (transcription)
- **Regulate**: Promoter/enhancer regulation
- **ThermoAttn**: Strand hybridization (cross-strand binding)
- **Epigenetic**: DNA methylation (persistent marks)
- **Allosteric**: Protein conformational signaling

9.1 Parameter Overhead

Per block, the additional parameters are:

- Epigenetic Memory: $\mathbf{M}_0(Md) + W_{\text{rk}}(d^2) + W_{\text{wk}}(d^2) + W_{\text{wv}}(d^2) + W_{\text{wg}}(d) + W_g(2d \times d) \approx 5d^2 + Md$
- Allosteric: $W_a(d) + W_s(d^2) + W_r(d^2) + W_g(2d \times d) \approx 4d^2$

For $d = 512$, $M = 256$: total $\approx 9 \times 512^2 + 256 \times 512 \approx$ 2.5M parameters per block—less than 10% overhead on top of the core DogmaBlock.

10 Related Work

Memory-augmented networks. NTM [6] and DNC [7] pioneered external memory with content-based addressing. Memory Networks [8] use end-to-end training for question answering. Our contribution is the biologically motivated gated write mechanism and integration with the DOGMA genomic computation paradigm. Critically, both NTM and DNC employ an external controller that generates read/write operations, whereas Epigenetic Memory integrates read/write directly into each layer's forward pass. This removes the controller bottleneck and enables the memory to participate in gradient flow through the full network depth.

Efficient long-context. Longformer [10], BigBird [11], and Landmark Attention [12] use sparse attention patterns. Mamba [15] uses selective state spaces. These approaches approximate full attention; our mechanisms provide *different functionality* (persistent memory and selective broadcast) rather than attention approximation. Notably, Mamba's selective gating mechanism shares conceptual similarity with our write gate g_t: both use input-dependent signals to control information flow. However, Mamba operates on a fixed-size hidden state per channel, whereas Epigenetic Memory maintains a structured bank of M content-addressable slots, enabling richer associative retrieval.

Linear-complexity recurrent models. RWKV [17] reformulates attention as a linear recurrence with time-decay, achieving $\mathcal{O}(Td)$ complexity and constant memory per step. RetNet [18] proposes a retention mechanism that unifies recurrent and parallel computation with multi-scale exponential decay. Both achieve linear complexity but rely on fixed decay schedules rather than content-based write gating. Our Epigenetic Memory provides *selective* persistence—the write gate g_t determines *what* to remember based on input content, whereas RWKV and RetNet apply uniform temporal decay regardless of information salience.

Modern Hopfield networks. Ramsauer et al. [21] show that transformer attention can be interpreted as the update rule of a modern continuous Hopfield network, where stored patterns are retrieved via energy minimization. Epigenetic Memory extends this perspective: the memory bank $\mathbf{M}$ acts as the stored pattern matrix, content-based read corresponds to Hopfield retrieval, and the gated write operation provides an *online* mechanism for updating stored patterns—a capability absent from classical Hopfield formulations.

Recurrent memory. Block-Recurrent Transformers [13] and Memorizing Transformers [9] add recur-

rent state. Epigenetic Memory provides analogous capability within a single forward pass without segment boundaries.

KV cache compression. Recent work on KV cache eviction policies [22] selectively retains "heavy hitter" tokens in the cache based on accumulated attention scores. While this addresses the memory cost of KV caches, it remains a heuristic compression of position-indexed activations. Epigenetic Memory provides an architecturally distinct solution: information is consolidated into a fixed-size content-addressable bank, avoiding the need for eviction heuristics entirely.

Biology-inspired architectures. Epigenetic mechanisms have been explored in evolutionary algorithms [16] but not as differentiable neural layers. Our work is the first to translate DNA methylation dynamics directly into a neural memory system.

11 Limitations and Future Work

Despite the strong theoretical properties and empirical results presented, several limitations warrant discussion.

Sequential memory interaction. The Epigenetic Memory read/write operations are inherently sequential (Algorithm 1, Phase I): each token must read from and write to the memory bank in order. This creates a sequential bottleneck of $\mathcal{O}(T)$ steps that cannot be fully parallelized, unlike the embarrassingly parallel computation of standard self-attention. For training, this can be partially mitigated through chunked processing (processing C tokens in parallel with intra-chunk independence assumptions), but the sequential dependency remains a practical limitation for hardware utilization on modern accelerators.

Memory slot interference. With soft addressing ($\boldsymbol{\beta}_t \in \Delta^{M-1}$), each write operation distributes its update across all M slots. When the memory is near capacity, new writes can destructively interfere with previously stored patterns—a phenomenon analogous to catastrophic forgetting in continual learning. While the contractive dynamics (Theorem 3) ensure convergence, they do not guarantee that all task-relevant patterns are preserved. Investigating memory consolidation strategies (e.g., periodic re-orthogonalization of $\mathbf{M}$) is a promising direction.

Top-k gradient discontinuity. The top-k selection in the Allosteric Regulation Network introduces a discontinuity in the gradient landscape: small changes in affinity scores a_i near the k-th threshold can cause discrete changes in the selected set $\mathcal{S}$. In practice, we use straight-through gradient estimation, but this is an approximation. Differentiable top-k relaxations (e.g., via the Gumbel-Softmax trick applied to subset selection) could provide smoother optimization and are left for future work.

Scaling to large models. Our experiments validate the approach at moderate scale ($d = 512$, 12 layers). The parameter overhead of $\sim$10% per block (Section 9) is modest, but the interaction between Epigenetic Memory and large-scale training dynamics (distributed optimization, mixed-precision training, gradient accumulation) remains unexplored. In particular, the memory bank $\mathbf{M}$ must be synchronized across data-parallel workers, introducing a potential communication overhead.

Biological fidelity vs. computational utility. Our biological analogies—DNA methylation for persistent memory and allosteric regulation for non-local communication—are deliberate simplifications. Real epigenetic systems involve multiple mark types (methylation, acetylation, phosphorylation) with complex combinatorial interactions. Extending the Epigenetic Memory Layer to support multiple mark types with cross-mark interactions could yield richer memory representations but at increased computational cost.

12 Conclusion

Biology solved the memory problem with two complementary mechanisms: epigenetic marks for persistent, selective storage, and allosteric regulation for efficient non-local communication. We translated both into differentiable neural layers and proved:

1. **Contractive dynamics** (Theorem 3): Memory converges to stable attractors.
2. **Capacity bounds** (Theorem 4): M slots store $\Theta(Md)$ bits of structured information.
3. **Information bottleneck** (Theorem 10): Top-k selection provides implicit regularization.
4. **Linear complexity** (Corollary 13): $\mathcal{O}(Td)$ for fixed M, k, versus $\mathcal{O}(T^2 d)$ for KV cache attention.

The KV cache remembers everything equally. Epigenetic memory remembers what matters. Allosteric regulation communicates what's important. Together, they provide biologically principled alternatives to brute-force quadratic attention for long-context sequence modeling.

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Algorithm 1 Epigenetic + Allosteric Forward Pass (per block)

Require: Input sequence $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_T] \in \mathbb{R}^{T \times d}$, memory bank $\mathbf{M} \in \mathbb{R}^{M \times d}$, top-$k$ parameter k
Ensure: Updated sequence $\hat{\mathbf{X}} \in \mathbb{R}^{T \times d}$, updated memory $\mathbf{M}'$

— **Phase I: Epigenetic Memory** —

for $t = 1, \ldots, T$ **do** ▷ Sequential memory interaction
  // Read: content-based retrieval
  $\mathbf{q}_t^r \leftarrow W_{\text{rk}}\, \mathbf{x}_t$ ▷ Read query
  $\boldsymbol{\alpha}_t \leftarrow \text{softmax}\big(\mathbf{q}_t^r \mathbf{M}^\top / \sqrt{d}\big)$ ▷ Read attention over M slots
  $\mathbf{m}_t \leftarrow \boldsymbol{\alpha}_t\, \mathbf{M}$ ▷ Retrieved memory vector
  // Gate: selective integration
  $\mathbf{x}_t' \leftarrow \mathbf{x}_t + \sigma\big(W_g[\mathbf{x}_t;\, \mathbf{m}_t]\big) \odot \mathbf{m}_t$ ▷ Gated residual
  // Write: selective gated deposit
  $\mathbf{k}_t^w \leftarrow W_{\text{wk}}\, \mathbf{x}_t; \quad \mathbf{v}_t^w \leftarrow W_{\text{wv}}\, \mathbf{x}_t$ ▷ Write key and value
  $g_t \leftarrow \sigma(W_{\text{wg}}\, \mathbf{x}_t)$ ▷ Write gate (scalar)
  $\boldsymbol{\beta}_t \leftarrow \text{softmax}\big(\mathbf{k}_t^w \mathbf{M}^\top / \sqrt{d}\big)$ ▷ Write address
  $\mathbf{M} \leftarrow \mathbf{M} + g_t \cdot (\boldsymbol{\beta}_t^\top \otimes \mathbf{v}_t^w)$ ▷ Memory update
end for

— **Phase II: Allosteric Regulation** —

  // Step 1: Compute binding affinities
for $i = 1, \ldots, T$ **do**
  $a_i \leftarrow W_a\, \mathbf{x}_i'$ ▷ Scalar affinity score
end for
  // Step 2: Select top-k binding sites
$\mathcal{S} \leftarrow \text{top-k}\big(\{a_i\}_{i=1}^T\big)$ ▷ $|\mathcal{S}| = k$
  // Step 3: Aggregate broadcast signal
$w_j \leftarrow e^{a_j} / \sum_{j' \in \mathcal{S}} e^{a_{j'}}$ for each $j \in \mathcal{S}$ ▷ Softmax over selected
$\mathbf{r} \leftarrow \sum_{j \in \mathcal{S}} w_j \cdot W_s\, \mathbf{x}_j'$ ▷ Weighted broadcast signal
  // Step 4: Gated reception at all positions
for $i = 1, \ldots, T$ **do**
  $\mathbf{c}_i \leftarrow W_r\, \mathbf{x}_i'$ ▷ Receptor embedding
  $g_i \leftarrow \sigma\big(W_g^{\text{allost}}[\mathbf{c}_i;\, \mathbf{r}]\big)$ ▷ Reception gate
  $\hat{\mathbf{x}}_i \leftarrow \mathbf{x}_i' + g_i \odot \mathbf{r}$ ▷ Final output
end for
$\mathbf{M}' \leftarrow \mathbf{M}$ ▷ Persist updated memory to next block/step
return $\hat{\mathbf{X}} = [\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_T]$, $\mathbf{M}'$